FAST COMPUTATION OF EHRHART POLYNOMIALS OF GELFAND–TSETLIN POLYTOPES VIA MACDONALD RECIPROCITY

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Abstract. We describe an efficient method for computing the Ehrhart polynomial of Gelfand–Tsetlin polytopes arising from Kostka coefficients. The key idea is to exploit Ehrhart–Macdonald reciprocity: evaluating the Ehrhart polynomial at negative integers reduces to counting *strict* Gelfand–Tsetlin patterns, which are often zero or very small for low dilations. Combined with an adaptive strategy that chooses the cheapest evaluation point (positive or negative) at each step, this yields substantial practical speedups compared to general-purpose polytope software. We benchmark against Oscar/polymake, and illustrate the broader applicability of the method through order polytopes and permutation posets.

1. Introduction

Let λ/μ be a skew shape and $\mathbf{w} = (w_1, \dots, w_k)$ a weight vector with $|\mathbf{w}| = |\lambda/\mu|$. The *Gelfand–Tsetlin polytope* $\mathrm{GT}(\lambda/\mu, \mathbf{w})$ is the set of all real-valued GT patterns of shape λ/μ and weight $\mathbf{w}$. Its integer points are in bijection with semistandard Young tableaux of shape λ/μ and weight $\mathbf{w}$, so

$$K_{\lambda/\mu,\mathbf{w}} = \#(\mathrm{GT}(\lambda/\mu, \mathbf{w}) \cap \mathbb{Z}^d), \tag{1}$$

where $d = \dim \mathrm{GT}(\lambda/\mu, \mathbf{w})$.

By a theorem of E. Rassart [Ras04], the function $n \mapsto K_{n\lambda/n\mu,n\mathbf{w}}$ is a polynomial in n: the *Ehrhart polynomial* $L_P(n)$ of the polytope $P = \mathrm{GT}(\lambda/\mu, \mathbf{w})$. Computing this polynomial is important for understanding the structure of Kostka coefficients and the geometry of GT polytopes.

The polynomiality of stretched Kostka and Littlewood–Richardson coefficients, and the structure of the resulting Ehrhart polynomials, have been studied by R. King, C. Tollu, and F. Toumazet [KTT04], who computed examples using direct enumeration.

Kostka coefficients arise as special cases of Littlewood–Richardson (LR) coefficients. By the identity $K_{\lambda/\nu,\mu} = \langle s_{\lambda/\nu}, h_\mu \rangle = \langle s_\lambda, s_\nu h_\mu \rangle$, we construct a skew shape ρ/σ that is a disjoint union (no shared rows or columns) of ν and single rows of lengths $\mu_1, \mu_2, \dots$, giving $s_{\rho/\sigma} = s_\nu h_\mu$ and hence $K_{\lambda/\nu,\mu} = c^{\rho}_{\lambda,\sigma}$. Thus software for LR coefficients such as A. Buch's `lrcalc` [Buc00] can be used for individual Kostka evaluations. However, the dilated partitions $n\lambda$, $n\mathbf{w}$ grow with n, making LR computation expensive at large dilations. Even `lrcalc` (highly optimized C code) takes over 8 minutes for the 11 evaluations needed for $\mathrm{GT}((3,3,3),(1^9))$ at dimension 10, compared to 1 ms for our dynamic programming (DP) approach with reciprocity (Table 2). Our horizontal-strip DP exploits the specific structure of Kostka coefficients directly, without the overhead of the LR reduction.

The standard approach evaluates $L_P(n)$ at $n = 1, 2, \ldots, d+1$ using a DP algorithm for Kostka coefficients, then recovers the polynomial by Lagrange interpolation. Each evaluation requires a DP that can be expensive for large dilations.

Our approach uses *Ehrhart–Macdonald reciprocity* to access the negative side of the polynomial for free or cheaply. The same idea also applies in other settings where one can count both lattice points and relative interior lattice points efficiently; in particular, we will later discuss order polytopes and an application to permutation posets.

2. Background

2.1. Ehrhart–Macdonald reciprocity. For a rational polytope $P \subset \mathbb{R}^d$ of dimension d (one whose vertices have rational coordinates), the function $L_P(n) = \#(nP \cap \mathbb{Z}^d)$ is a quasi-polynomial in n, and a polynomial when P is a lattice polytope. GT polytopes are rational but in general not lattice polytopes, so one would *a priori* expect L_P to be only a quasi-polynomial. However, by Rassart's theorem [Ras04], L_P is a genuine polynomial—a case of so-called *period collapse*. Ehrhart–Macdonald reciprocity [Mac71, BR15] gives:

$$(-1)^d L_P(-n) = L_P^*(n) \coloneqq \#(\mathrm{int}(nP) \cap \mathbb{Z}^d), \tag{2}$$

where $\mathrm{int}(nP)$ denotes the relative interior of nP.

2.2. Gelfand–Tsetlin polytopes. Recall that an SSYT of skew shape λ/μ and content $\mathbf{w} = (w_1, \ldots, w_k)$ corresponds bijectively to a chain of partitions

$$\mu = \alpha^{(0)} \subseteq \alpha^{(1)} \subseteq \cdots \subseteq \alpha^{(k)} = \lambda,$$

where each $\alpha^{(i)}/\alpha^{(i-1)}$ is a horizontal strip of size w_i. The *Gelfand–Tsetlin polytope* $\mathrm{GT}(\lambda/\mu, \mathbf{w}) \subset \mathbb{R}^m$ is the convex polytope obtained by relaxing the integrality constraint: the entries of the intermediate partitions $\alpha^{(1)}, \ldots, \alpha^{(k-1)}$ are allowed to be real, subject to the interlacing inequalities

$$\alpha_j^{(i-1)} \leq \alpha_j^{(i)} \leq \alpha_{j-1}^{(i-1)}$$

and the row-sum constraints $|\alpha^{(i)}| - |\alpha^{(i-1)}| = w_i$. The lattice points of $\mathrm{GT}(\lambda/\mu, \mathbf{w})$ are precisely the SSYT, so $K_{\lambda/\mu,\mathbf{w}} = \#(\mathrm{GT}(\lambda/\mu, \mathbf{w}) \cap \mathbb{Z}^m)$.

2.3. Strict GT patterns. The interior lattice points of $n \cdot \mathrm{GT}(\lambda/\mu, \mathbf{w})$ are precisely the *strict GT patterns*: those where every interlacing inequality that is *not* forced as an equality in the affine hull of the polytope is strict. Equivalently, one imposes strictness only on the non-globally-tight inequalities. Thus

$$L_P^*(n) = K^{\mathrm{strict}}_{n\lambda/n\mu, n\mathbf{w}}, \tag{3}$$

where K^{strict} counts strict semistandard tableaux (equivalently, GT patterns with strictness on all non-forced internal inequalities). This is computed by the same DP as the ordinary Kostka coefficient, but with strict inequality constraints.

3. The adaptive reciprocity algorithm

To compute $L_P(n)$ for a polytope of dimension d, we need $d+1$ evaluations at distinct integers. We always have $L_P(0) = 1$ for free, and by (2) and (3), each negative evaluation $(-1)^d L_P(-n) = K^{\text{strict}}_{n\lambda,n\mu,n\mathbf{w}}$ reduces to a strict Kostka computation. The key observation is that strict Kostka coefficients are often *zero* for small n: if any part of $\mathbf{w}$ is 1, then strict GT patterns require the corresponding row to have distinct entries in a range that may be empty for small dilations. Even when the strict count is nonzero, its DP state space is typically much smaller than the ordinary Kostka DP for the same dilation.

This motivates an adaptive strategy:

Input: Shape λ/μ, weight $\mathbf{w}$, dimension d
Output: Ehrhart polynomial $L_P(n)$
Initialize: points $\leftarrow \{(0,1)\}$;
$p \leftarrow 0$, $q \leftarrow 0$;
$\text{cost}_+ \leftarrow 1$, $\text{cost}_- \leftarrow 1$;
while $|points| \le d$ **do**
  if $cost_- \le cost_+$ **then**
    $q \leftarrow q+1$;
    $v \leftarrow K^{\text{strict}}_{q\lambda,q\mu,q\mathbf{w}}$;
    Append $(-q,\ (-1)^d v)$ to points;
    $\text{cost}_- \leftarrow v$;
  else
    $p \leftarrow p+1$;
    $v \leftarrow K_{p\lambda,p\mu,p\mathbf{w}}$;
    Append (p, v) to points;
    $\text{cost}_+ \leftarrow v$;
  end
end
return Lagrange interpolation of points;

Algorithm 1: Adaptive Ehrhart polynomial computation

The "cost" heuristic uses the most recent Kostka/strict-Kostka value as a proxy for the DP state-space size at the next dilation. The negative side typically wins for the first several points because strict Kostka equals zero.

Remark 3.1. The negative evaluations are completely free when $K^{\text{strict}} = 0$. For a weight vector with k ones, the strict Kostka coefficient $K^{\text{strict}}_{n\lambda,n\mu,n\mathbf{w}}$ is zero whenever n is small enough that the strict interlacing constraints for those unit-weight rows cannot be satisfied. In practice, half or more of the $d+1$ required evaluation points are obtained this way.

4. Implementation

Our implementation is in Rust, available as the `kostka` crate.[1] At its core is a DP algorithm for (skew) Kostka coefficients $K_{\lambda/\mu,\mathbf{w}}$ based on horizontal-strip removal, using `BigUint` arithmetic for exact computation. A variant with strict inequality constraints in the DP transitions handles the strict Kostka coefficients $K^{\text{strict}}_{\lambda/\mu,\mathbf{w}}$. The dimension of $\mathrm{GT}(\lambda/\mu,\mathbf{w})$ is determined by counting free entries in the GT pattern with propagation of forced values; see J. A. De Loera and T. B. McAllister [DM06] for related degree computations of stretched Kostka coefficients. The Ehrhart polynomial is then recovered from evaluation points via Lagrange interpolation over $\mathbb{Q}$.

The `kostka` CLI supports:

```
kostka ehrhart --lambda 3,2,1 -w 1,1,1,1,1,1
kostka ehrhart --lambda 4,3 --mu 2,1 -w 2,2,2
kostka hstar --lambda 4,3,2,1 -w 1,1,1,1,1,1,1,1,1,1
kostka hstar --lambda 4,3,1 --mu 2,1 -w 2,2,1
```

The tool also computes Kostka coefficients associated with flagged Schur functions, Littlewood–Richardson coefficients, and batch computations over all weights for a given shape. We refer to the Git repository for full documentation.

5. Benchmarks

We compare our method against two alternatives: the Oscar/polymake pipeline [GJ00, Hog25] and A. Buch's `lrcalc` [Buc00], which reduces Kostka evaluations to LR coefficient computations. (Oscar is a Julia framework that calls polymake for polyhedral computations, which in turn delegates lattice-point enumeration and h^*-vector computation to Normaliz [BI10]; thus "Oscar," "polymake," and "Normaliz" refer to different layers of the same computational stack.) Tables 1 and 2 summarize the results.

The Oscar/polymake approach requires constructing the full vertex representation of the GT polytope—expensive in high dimensions—and does not exploit Ehrhart–Macdonald reciprocity. Our method avoids vertex enumeration entirely: each evaluation of $L_P(n)$ or $L^*_P(n)$ reduces to a Kostka coefficient DP whose complexity depends on the dilation n (not the number of vertices), and reciprocity provides many free evaluation points where strict Kostka equals zero.

Remark 5.1. The observed wall-clock ratios are typically between 10^4 and 10^5. The Oscar/polymake timings are from [Hog25], computed on a laptop using the Julia Oscar framework (which calls polymake/Normaliz internally for h^*-vector computation). The total computation time reported in [Hog25]

[1] `https://github.com/PerAlexandersson/kostka`

TABLE 1. Comparison of Ehrhart polynomial computation times. "d" is the polytope dimension. Column "`kostka`" is our DP + reciprocity method (desktop). Column "Oscar" is from [Hog25] using Oscar/polymake (laptop; because the hardware differs, the reported ratios should be read as indicative rather than exact speedups). Cases marked "—" exceeded the available time/memory.

λ	$\mathbf{w}$	d	`kostka`	Oscar	speedup
Direct comparisons (partitions of 10, same cases):					
$(8,1,1)$	$(2,1^8)$	12	2 ms	666 s	3×10^5
$(4,4,2)$	$(2,1^8)$	12	2 ms	182 s	9×10^4
$(4,3,3)$	$(2,1^8)$	12	2 ms	230 s	10^5
$(5,3,2)$	$(2,1^8)$	13	4 ms	764 s	2×10^5
$(7,2,1)$	$(2,1^8)$	13	3 ms	1347 s	4×10^5
$(7,1,1,1)$	$(2,1^8)$	15	5 ms	467 s	9×10^4
$(4,2,2,2)$	$(2,1^8)$	15	9 ms	125 s	10^4
$(3,3,3,1)$	$(2,1^8)$	15	8 ms	573 s	7×10^4
$(5,2,2,1)$	$(2,1^8)$	17	21 ms	1988 s	9×10^4
$(5,3,1,1)$	$(2,1^8)$	17	18 ms	1179 s	7×10^4
Cases beyond Oscar's reach:					
$(4,3,2,1)$	(1^{10})	21	104 ms	—	
$(5,5,5)$	(1^{15})	22	45 ms	—	
$(5,3,3,1,1,1)$	$(2^4,1^6)$	26	17 s	—	

for all partitions of size 10 was over 412 hours. Our method computes the same cases in under 1 second total, and extends to much larger cases (55,000 polytopes up to dimension 26 in our database).

For standard weight $\mathbf{w}=(1^N)$, the reciprocity method is especially effective: $K^{\text{strict}}_{n\lambda,n\mathbf{w}}=0$ for all small n, so many evaluation points are free. For example, computing the degree-21 Ehrhart polynomial for $\mathrm{GT}((4,3,2,1),(1^{10}))$ requires 22 evaluation points, of which the first 9 negative points all give $K^{\text{strict}}=0$.

Remark 5.2. The slowdown of `lrcalc` at higher dilations is expected: computing $c^{n\rho}_{n\lambda,n\sigma}$ requires enumerating Littlewood–Richardson tableaux of shape $n\rho/n\lambda$ with content $n\sigma$, whose count grows rapidly with n. Our DP, by contrast, processes one row of the GT pattern at a time and benefits from the horizontal-strip structure. The reciprocity advantage (free evaluations at negative integers) further widens the gap.

6. THE h^*-VECTOR

For a *rational* polytope P with denominator q (the smallest positive integer such that qP is a lattice polytope), the lattice-point counting function

TABLE 2. Comparison with `lrcalc` [Buc00]. Each Kostka evaluation $K_{n\lambda,n\mathbf{w}}$ is computed via the Stanley reduction to an LR coefficient $c^{n\rho}_{n\lambda,n\sigma}$ and then calling `lrcalc lrcoef`. Both methods run on the same desktop. The `lrcalc` column is the total time for all $d+1$ positive-dilation evaluations (no reciprocity).

λ	$\mathbf{w}$	d	`kostka`	`lrcalc`	speedup
$(3,2,1)$	(1^6)	7	<1 ms	86 ms	$\sim 10^2$
$(3,3,3)$	(1^9)	10	1 ms	491 s	$\sim 6\times 10^3$
$(4,3,2)$	(1^9)	13	4 ms	>10 min	$>10^5$

$L_P(n) = |nP \cap \mathbb{Z}^d|$ is a *quasi-polynomial* of degree d and period dividing q. The Ehrhart series then satisfies

$$\sum_{n\geq 0} L_P(n)\, z^n = \frac{h^*_q(P;z)}{(1-z^q)^{d+1}}, \tag{4}$$

where $h^*_q(P;z)$ is a polynomial of degree at most $q(d+1)-1$ with non-negative integer coefficients, see [BS07] and [Sta93]. Since GT polytopes have period collapse by Rassart's theorem, we define the *h^*-vector* $(h^*_0,\dots,h^*_d)$ as in the lattice polytope case ($q=1$):

$$\sum_{n\geq 0} L_P(n)\, z^n = \frac{h^*_0 + h^*_1 z + \cdots + h^*_d z^d}{(1-z)^{d+1}}. \tag{5}$$

Comparing (4) and (5) via the factorization $(1-z^q)^{d+1} = (1-z)^{d+1}(1+z+\cdots+z^{q-1})^{d+1}$, we see that $h^*_q(P;z) = (h^*_0+\cdots+h^*_d z^d)\cdot(1+z+\cdots+z^{q-1})^{d+1}$. In particular, non-negativity of the h^*_i implies non-negativity of the coefficients of $h^*_q(P;z)$. R. King, C. Tollu, and F. Toumazet [KTT04] conjectured that the Ehrhart polynomial $L_P(n)$ of a GT polytope has non-negative coefficients (in the standard monomial basis). Note that this is an independent property from h^*-positivity: neither implies the other in general. Our computations verify both properties for all straight-shape GT polytopes $\mathrm{GT}(\lambda,(1^{|\lambda|}))$ with $|\lambda|\leq 15$—over 54,000 cases—providing further evidence for the conjecture. See also [AA19, AO23] for related positivity conjectures for Ehrhart polynomials of polytopes related to key polynomials, flagged Schur functions, and cylindric Schur functions.

As observed by C. Haase and T. B. McAllister [HM08], there exist rational polytopes with period collapse whose Ehrhart polynomial is not the Ehrhart polynomial of any lattice polytope. It is natural to ask whether the same phenomenon occurs for GT polytopes: is $L_{\mathrm{GT}(\lambda/\mu,\mathbf{w})}(n)$ always realizable as the Ehrhart polynomial of a lattice polytope, or are GT polytopes genuinely different in this respect?

7. Order polytopes

The reciprocity approach extends naturally to *order polytopes*. For a finite poset P on $\{0, 1, \ldots, n-1\}$, the *order polytope* [Sta86]

$$\mathcal{O}(P) = \{x \in [0,1]^n : x_a \leq x_b \text{ for all } a <_P b\}$$

has the property that its lattice points at dilation t are weakly order-preserving maps $P \to \{0, 1, \ldots, t\}$, so $L_{\mathcal{O}(P)}(t) = \Omega(P, t+1)$, where $\Omega(P, k)$ is Stanley's order polynomial. Interior lattice points at dilation t correspond to strictly order-preserving maps $P \to \{1, \ldots, t-1\}$, giving Ehrhart–Macdonald reciprocity in the form

$$L_{\mathcal{O}(P)}(-k) = (-1)^n \, \bar{\Omega}(P, k-1), \tag{6}$$

where $\bar{\Omega}(P, m)$ counts strict order-preserving maps $P \to \{1, \ldots, m\}$.

7.1. Frontier DP for order-preserving maps.

The standard approach to evaluating $\Omega(P, k)$ enumerates all order-preserving maps by backtracking over vertices, giving $O(k^n)$ worst-case time. We introduce a *frontier DP* that processes vertices in a natural labeling order, maintaining as state only the values of *live* vertices—those with at least one unprocessed child. When a vertex v has no children among the remaining vertices, its value does not affect future constraints, so all valid choices are collapsed into a single multiplier $(k - \ell_v + 1)$, where ℓ_v is the lower bound from parent constraints. States that agree on all live vertex values are merged.

The complexity is $O(n \cdot k^w)$, where w is the maximum frontier width (the largest number of simultaneously live vertices). For chains and zigzag (fence) posets, $w = 1$; for antichains, $w = 0$ (each vertex is independent); for Young diagram posets of shape $(\lambda_1, \ldots, \lambda_r)$, $w \leq \lambda_1$.

7.2. Order polytope benchmarks.

We benchmark three methods for computing the h^*-vector of order polytopes:

(1) **Rust DP**: our frontier DP with Ehrhart–Macdonald reciprocity and `BigRational` interpolation;
(2) **linext**: Stanley's formula—enumerate all linear extensions $\mathcal{L}(P)$ and tabulate descents (streaming, $O(n)$ memory);
(3) **polymake**: the polymake/Normaliz backend for general lattice-point enumeration.

Remark 7.1. Table 3 reveals that the three methods have complementary strengths. For *few linear extensions* (shapes with small hook lengths), Stanley's formula is fastest: the staircase $(4, 3, 2, 1)$ has only 768 linear extensions, so direct enumeration takes 30 µs. For *narrow* posets (chains, fences), the frontier DP dominates because $w = 1$ gives polynomial time $O(nk)$, while the number of linear extensions grows exponentially—fence(14) already exhausts the 5 s cap for polymake, but the DP finishes in 6 ms. Polymake/Normaliz is a general-purpose tool that does not exploit the order polytope structure; its triangulation-based approach scales with the number

TABLE 3. Comparison of order polytope h^*-vector computation times. All three methods run on the same machine (desktop, single thread). "—" indicates the method was not run (the linear extension method is infeasible for large fence posets due to the exponential number of extensions).

Poset	n	Rust DP	linext	polymake
Shape $(3, 2, 1)$	6	0.4 ms	2 µs	60 ms
Fence(8)	8	0.9 ms	50 µs	90 ms
Shape $(4, 3, 2, 1)$	10	2 ms	30 µs	90 ms
Shape $(3, 3, 3)$	9	1.3 ms	5 µs	60 ms
Fence(10)	10	1.8 ms	1.7 ms	520 ms
2-alt(10)	10	1.8 ms	1.7 ms	490 ms
Shape $(5, 4, 3, 2, 1)$	15	9 ms	11 ms	2.5 s
Shape $(4, 4, 4, 4)$	16	20 ms	2 ms	270 ms
Shape $(5, 5, 5)$	15	20 ms	0.4 ms	120 ms
Fence(14)	14	6 ms	—	> 5 s
Shape $(5, 5, 5, 5)$	20	175 ms	—	> 5 s
Shape $(6, 5, 4, 3, 2, 1)$	21	100 ms	—	—
Fence(20)	20	23 ms	—	—
Shape $(6, 6, 6)$	18	230 ms	—	—
Shape $(6, 6, 6, 6)$	24	6.7 s	—	—

of vertices of the polytope, not the poset width. For the largest cases ($n \geq 20$), only the frontier DP is feasible.

By Stanley's theorem [Sta11, Thm. 4.5.14], for a naturally labeled poset P, the h^*-vector of $\mathcal{O}(P)$ equals the P-Eulerian polynomial:

$$h_i^* = \#\{\sigma \in \mathcal{L}(P) : \mathrm{des}(\sigma) = i\},$$

where $\mathcal{L}(P)$ is the set of linear extensions and $\mathrm{des}(\sigma)$ counts descents. Counting linear extensions is $\#\mathsf{P}$-complete in general [BW91], and even for structured posets the number $|\mathcal{L}(P)|$ can be enormous—e.g., $|\mathcal{L}(P)| = n!$ for the antichain on n elements. Our Ehrhart/reciprocity approach avoids linear extension enumeration entirely: it requires only $\sim n/2$ evaluations of the order polynomial via the frontier DP, each costing $O(n \cdot k^w)$.

Example 7.2. For the fence (zigzag) poset on 10 elements:

$$\mathbf{h}^* = (1, 133, 2475, 12331, 20641, 12331, 2475, 133, 1).$$

This is palindromic, as expected for a Gorenstein polytope.

7.3. **Non-real-rooted h^*-vectors from permutation posets.** As an application of the fast h^*-computation, we investigate real-rootedness of h^*-polynomials for *permutation posets*. Given $w \in S_n$, the permutation poset P_w is the partial order on $\{1, \dots, n\}$ with $i <_{P_w} j$ iff $i < j$ and $w_i < w_j$.

Since P_w is always naturally labeled, $h^*(\mathcal{O}(P_w); t)$ equals the P-Eulerian polynomial by Stanley's theorem.

Starting from a 321-avoiding permutation of Stembridge,

$$w_0 = (2, 4, 6, 8, 10, 1, 12, 3, 15, 5, 17, 7, 9, 11, 13, 14, 16) \in S_{17},$$

whose permutation poset has $h^*(t) = 1+32t+336t^2+1420t^3+2534t^4+1946t^5+658t^6+86t^7+3t^8$ (ultra-log-concave but *not* real-rooted), we searched over 34,226 nearby 4321-avoiding permutations (within 3 arbitrary transpositions) and found exactly 22 with non-real-rooted h^*.

Example 7.3. Almost all non-real-rooted examples are 321-avoiding. A notable exception, found at transposition distance 3 from w_0, is

$$w = (3, 4, 6, 8, 10, \underline{12, 2, 1}, 15, 5, 17, 7, 9, 11, 13, 14, 16) \in S_{17},$$

which *contains* the pattern 321. Its h^*-polynomial is

$$h^*(t) = 1 + 41t + 525t^2 + 2596t^3 + 5349t^4 + 4731t^5 + 1849t^6 + 284t^7 + 12t^8,$$

with non-real roots at approximately $-2.15 \pm 0.041\, i$. This is the only 321-containing example among the 22 failures at $n = 17$, and has the *smallest* imaginary part of all examples found—it is the closest to being real-rooted. The coefficients form an ultra-log-concave sequence.

8. Further directions

Our method applies whenever one has a combinatorial interpretation of both $L_P(n)$ and the interior count $L_P^*(n)$ that admits a DP or transfer-matrix algorithm, together with a regime where $L_P^*(n) = 0$ for small n. GT polytopes are closely related to *flow polytopes* on directed acyclic graphs [LMD19].

Matroid polytopes are another natural family where we believe the reciprocity approach will be effective.

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